

МИНОБРНАУКИ РОССИИ

Федеральное государственное бюджетное образовательное учреждение

высшего образования

«МИРЭА - Российский технологический университет»

РТУ МИРЭА

Институт Информационных Технологий

Кафедра Математического обеспечения и

стандартизации информационных технологий (МОСИТ)

ОТЧЁТ ПО ПРАКТИЧЕСКОЙ РАБОТЕ № 1

по дисциплине

«Виртуализация вычислительной инфраструктуры»

Выполнил студент группы *Лияскин.В.А*

ИВБО-20-24

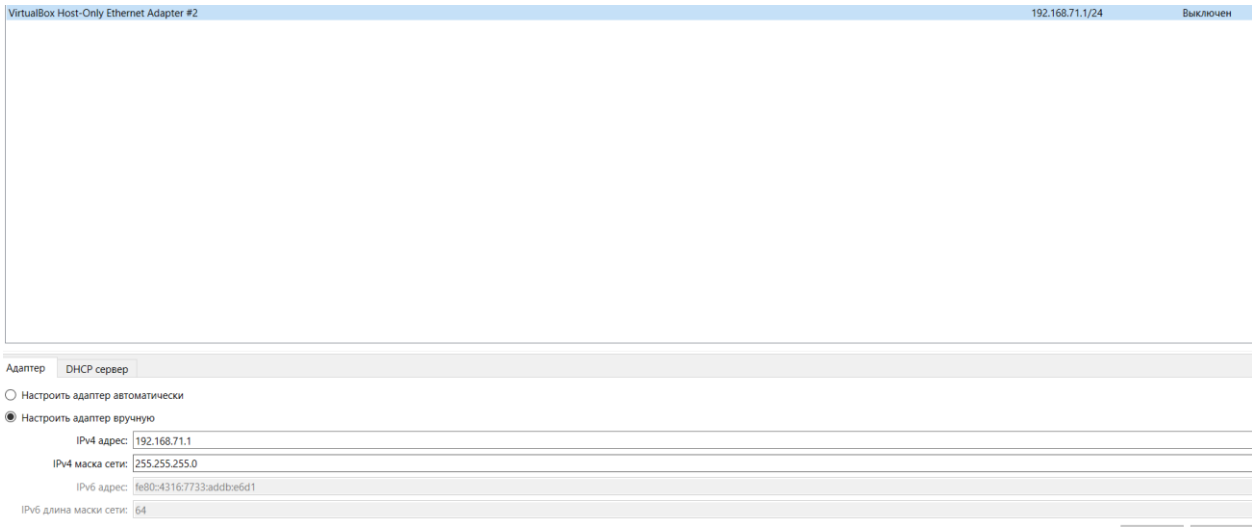
Принял *Воронцов.Ю.А*

Практическая работа выполнена « __ » _____ 2026 г.

«Зачтено» « _ » _____ 2026 г.

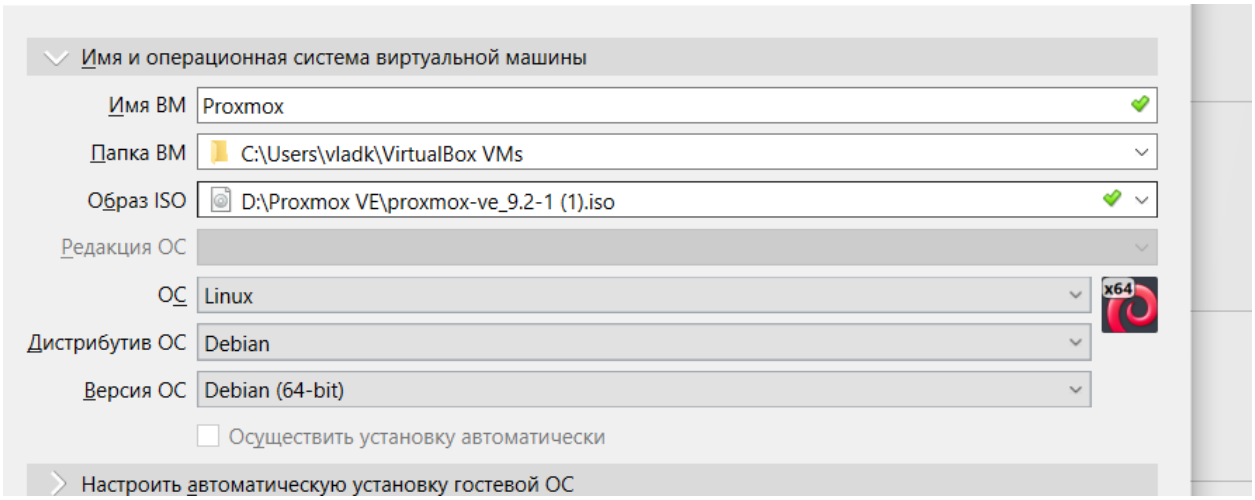
Москва 2026

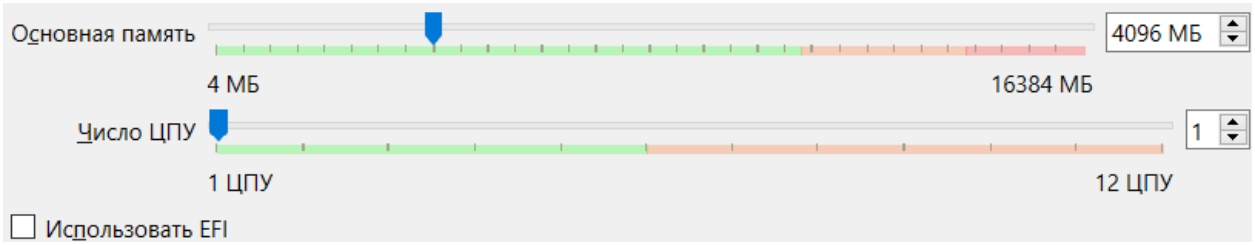
Шаг 1: Подготовка сети в VirtualBox

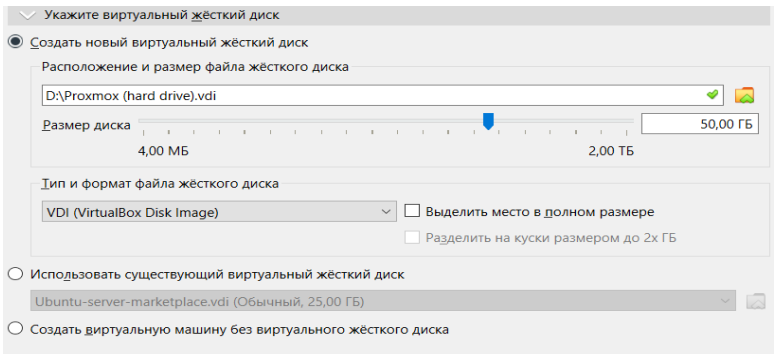


Подсеть – 192.168.71.0/24, шлюз – 192.168.71.1

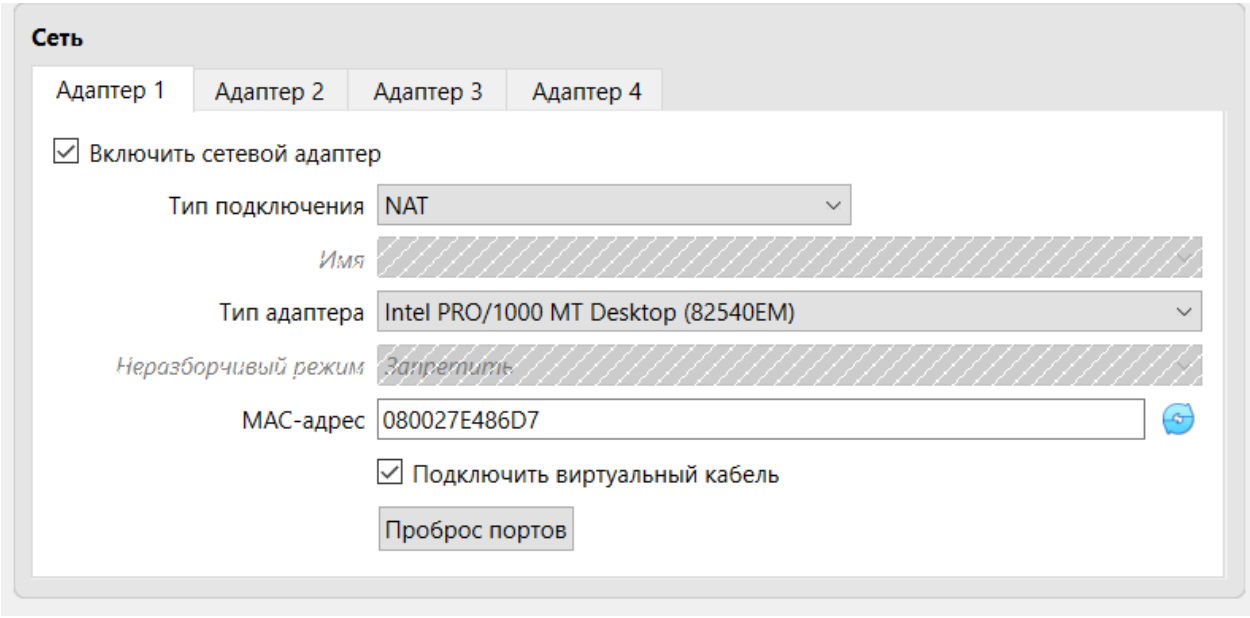
Шаг 2: Создание виртуальной машины

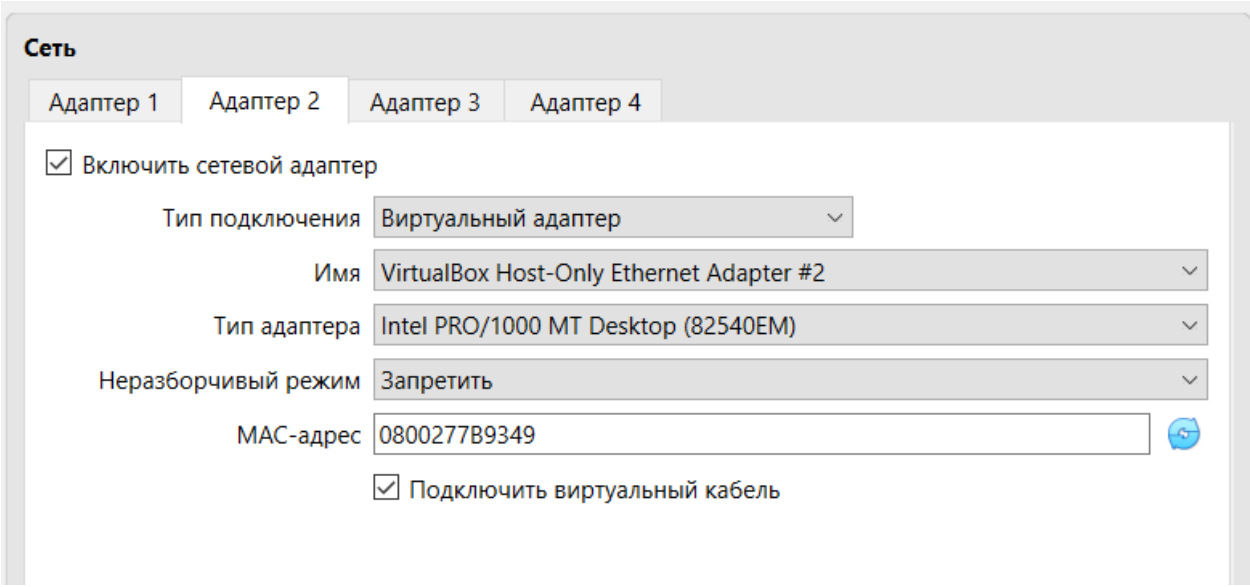






Шаг 3: Настройка сетевых адаптеров



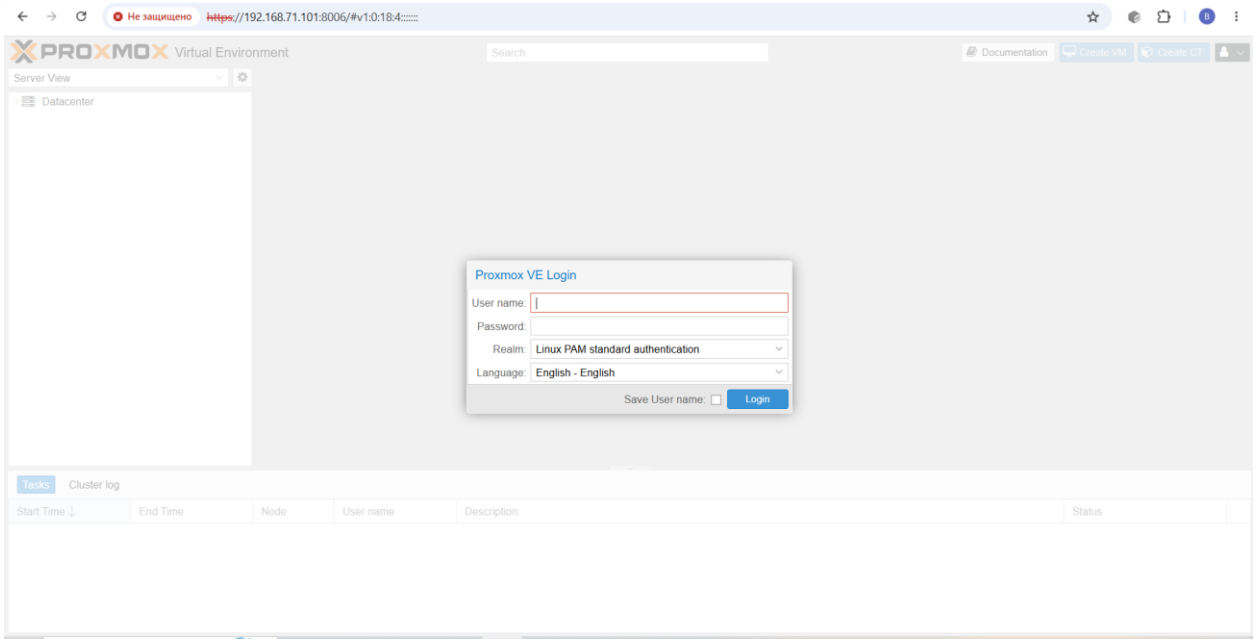


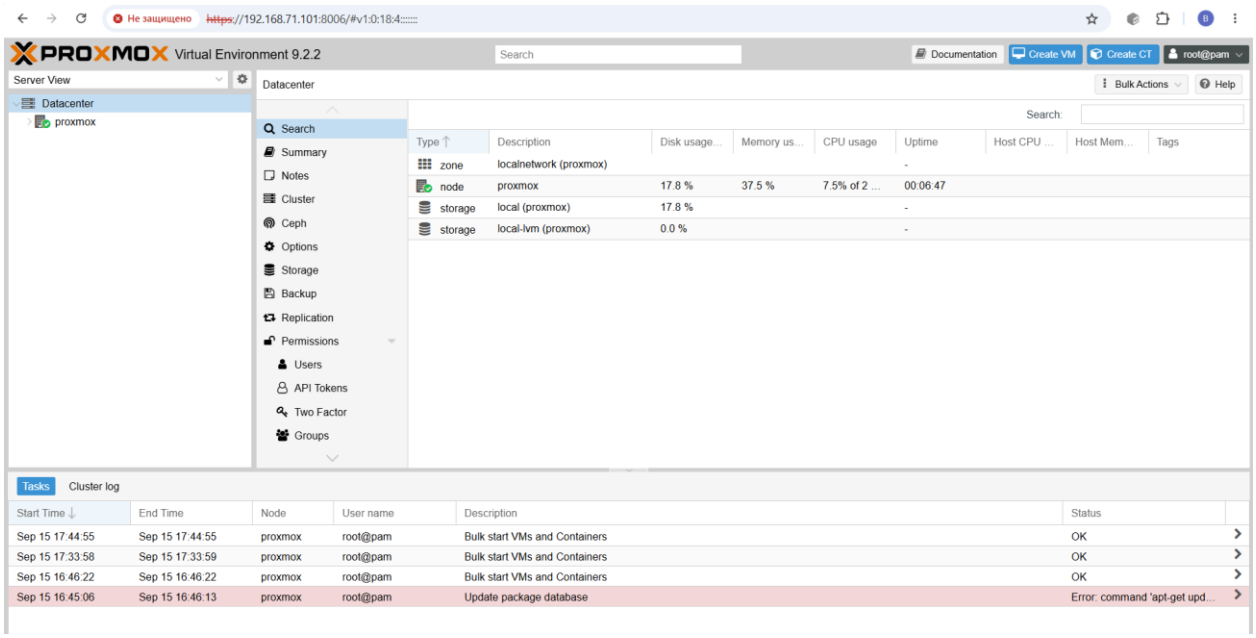
Шаг 4: Установка Proxmox VE



ЭТАП 2: Вход в веб-интерфейс и настройка сети

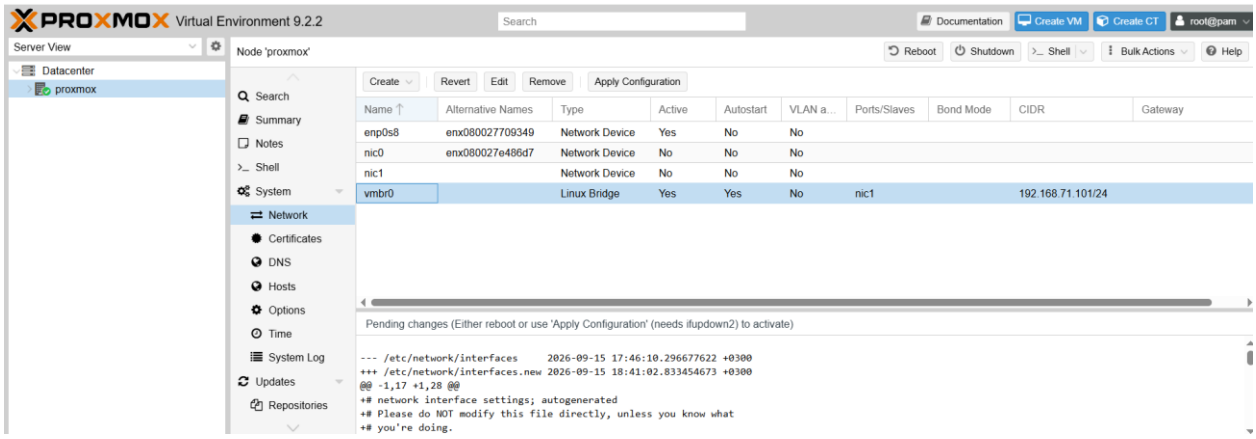
Переходим по адресу на физической машине и попадаем в настройку proxmox



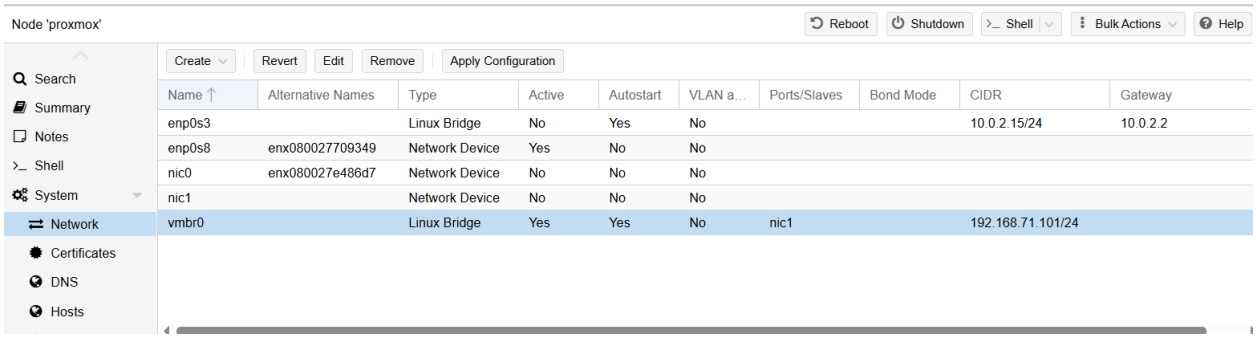


Настройка интерфейса enp0s3 (NAT)

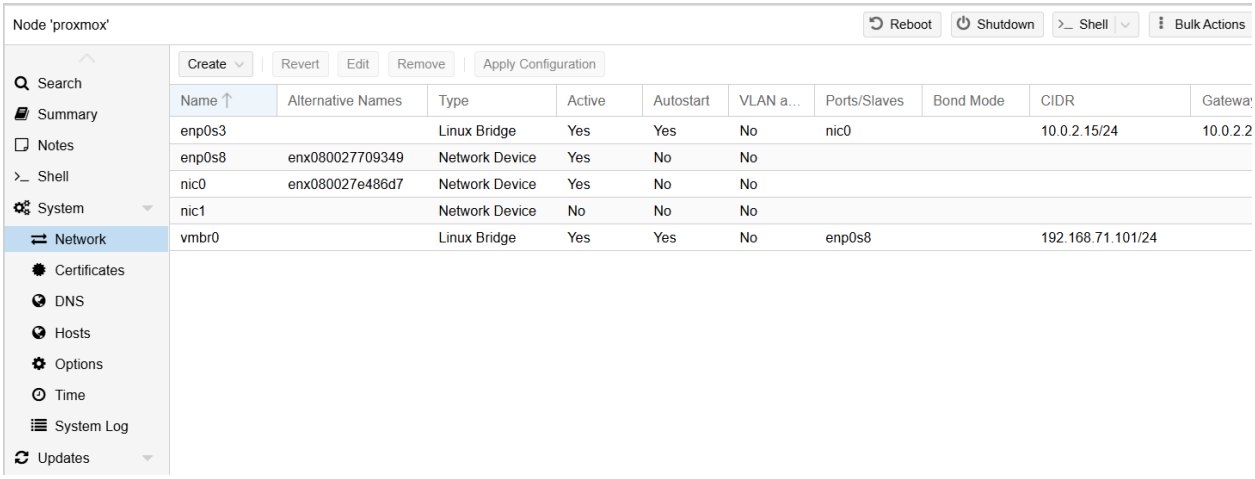
Удаляю gateway у vmbr0



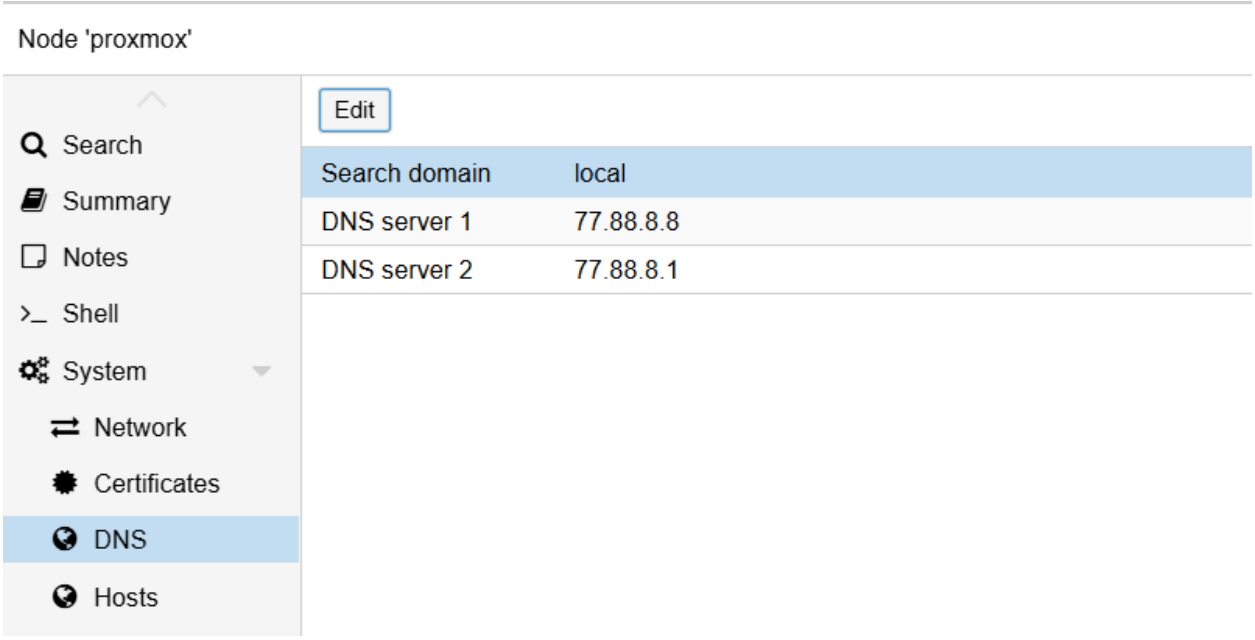
Создание интерфейса для NAT



Применение конфигурации



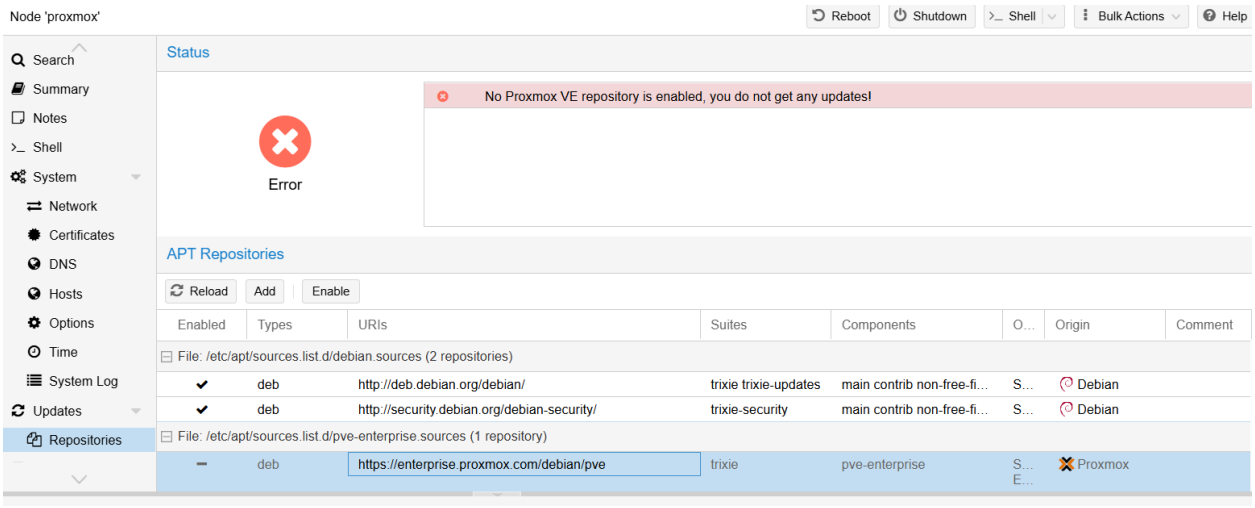
Настройка DNS



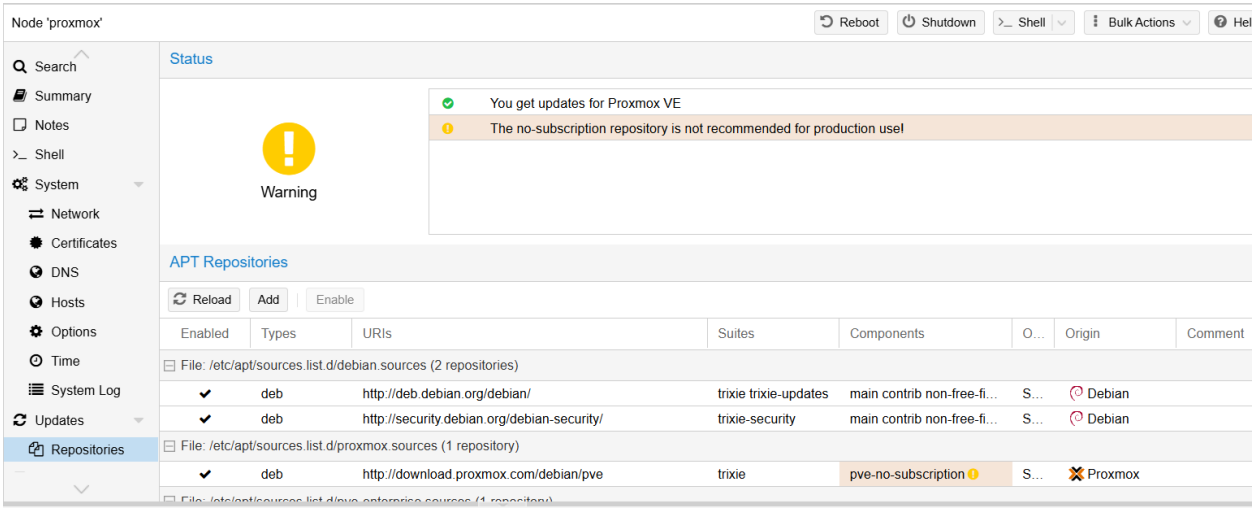
И проверка

```
root@proxmox:~# cat /etc/resolv.conf
search local
nameserver 77.88.8.8
nameserver 77.88.8.1
root@proxmox:~# ping ya.ru
PING ya.ru (5.255.255.242) 56(84) bytes of data.
64 bytes from ya.ru (5.255.255.242): icmp_seq=1 ttl=255 time=8.79 ms
64 bytes from ya.ru (5.255.255.242): icmp_seq=2 ttl=255 time=10.0 ms
64 bytes from ya.ru (5.255.255.242): icmp_seq=3 ttl=255 time=18.4 ms
64 bytes from ya.ru (5.255.255.242): icmp_seq=4 ttl=255 time=11.2 ms
^C
--- ya.ru ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3006ms
rtt min/avg/max/mdev = 8.789/12.118/18.408/3.732 ms
root@proxmox:~#
```

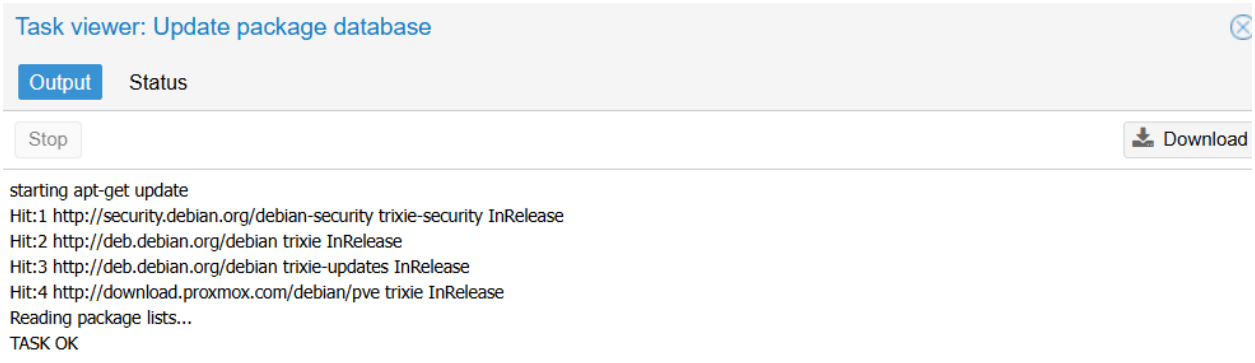
Отключение платного репозитория



Добавление бесплатного репозитория



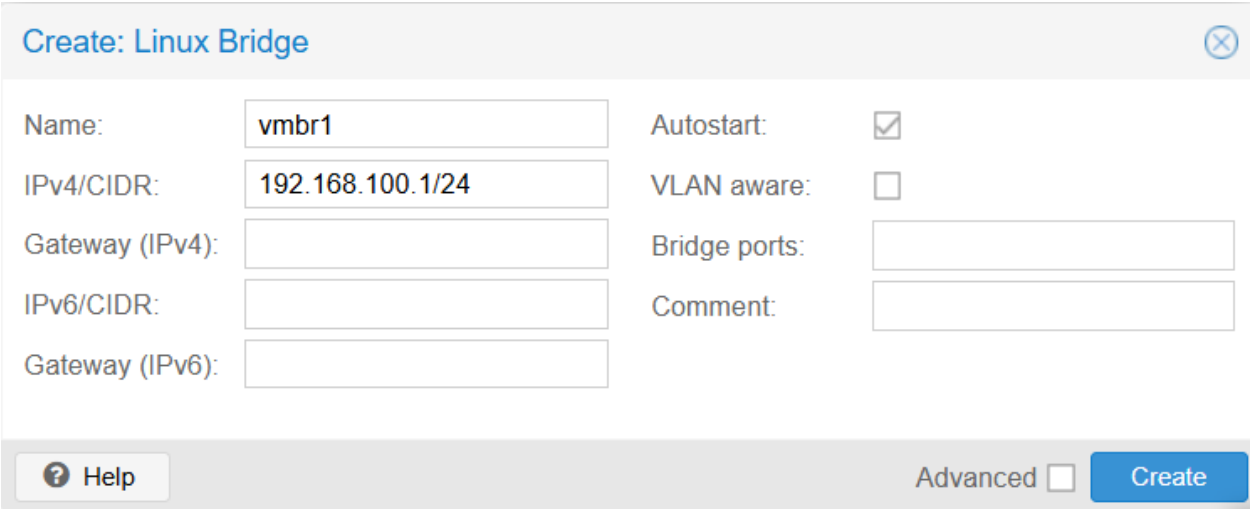
Обновление системы



Проверяем, что все обновлено

```
root@proxmox:~# apt update && apt upgrade
Hit:1 http://deb.debian.org/debian trixie InRelease
Hit:2 http://deb.debian.org/debian trixie-updates InRelease
Hit:3 http://security.debian.org/debian-security trixie-security InRelease
Hit:4 http://download.proxmox.com/debian/pve trixie InRelease
All packages are up to date.
Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 0
```

Создание моста vmbr1 для гостевых ВМ



Настройка NAT для vmbr1

```
iface vmbr1 inet static
        address 192.168.100.1/24
        bridge-ports none
        bridge-stp off
        bridge-fd 0
        post-up echo 1 > /proc/sys/net/ipv4/ip_forward
        post-up iptables -t nat -A POSTROUTING -s '192.168.100.0/24' -o enp0s3 -j MASQUERADE
        post-down iptables -t nat -D POSTROUTING -s '192.168.100.0/24' -o enp0s3 -j MASQUERADE
source /etc/network/interfaces.d/*
```

Проверяем правила NAT

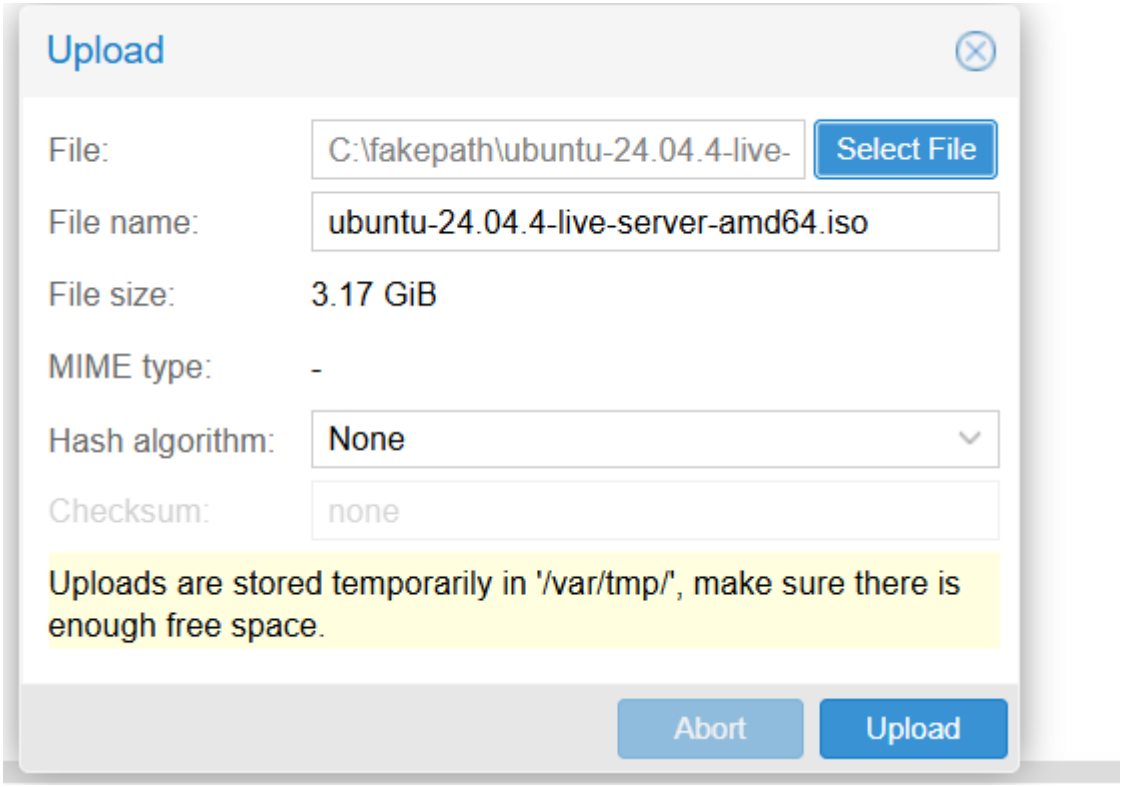
```
root@proxmox:~# iptables -t nat -L POSTROUTING -n -v
Chain POSTROUTING (policy ACCEPT 28 packets, 2140 bytes)
 pkts bytes target     prot opt in     out     source               destination
    0     0 MASQUERADE  all  --  *      enp0s3  192.168.100.0/24     0.0.0.0/0
```

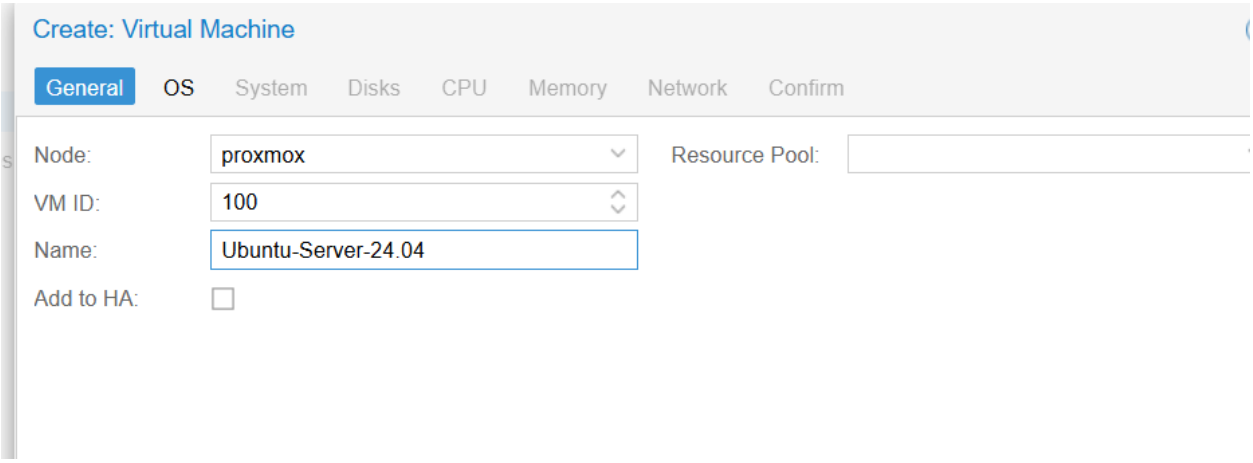
Задание №4 Создание виртуальной машины

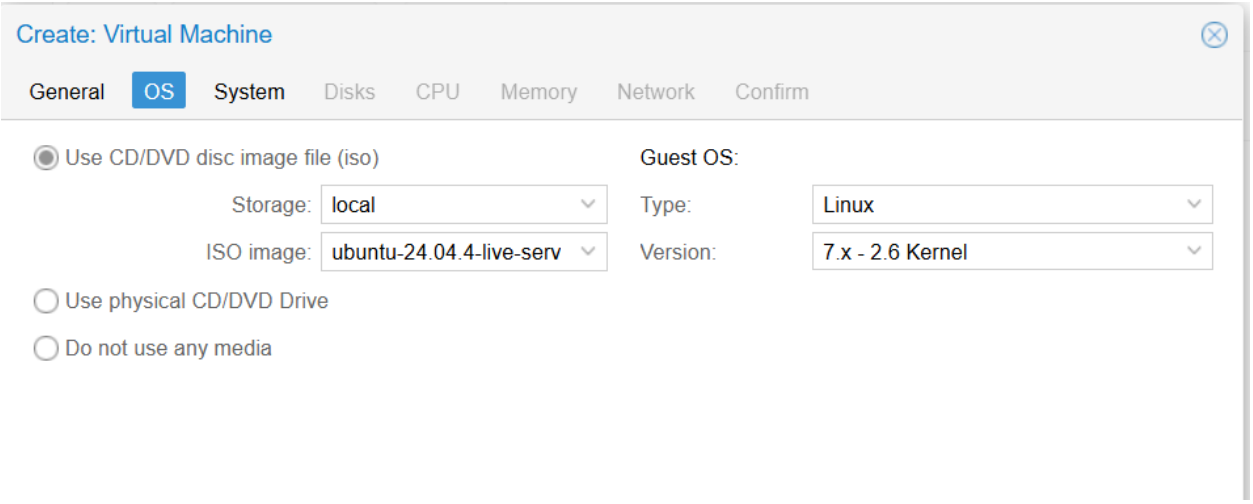
1.Скачиваем файл ISO

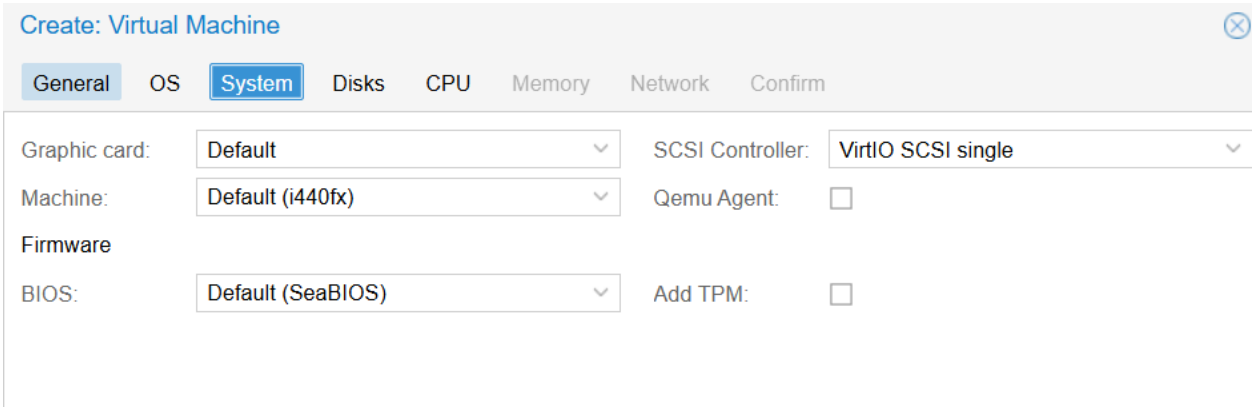


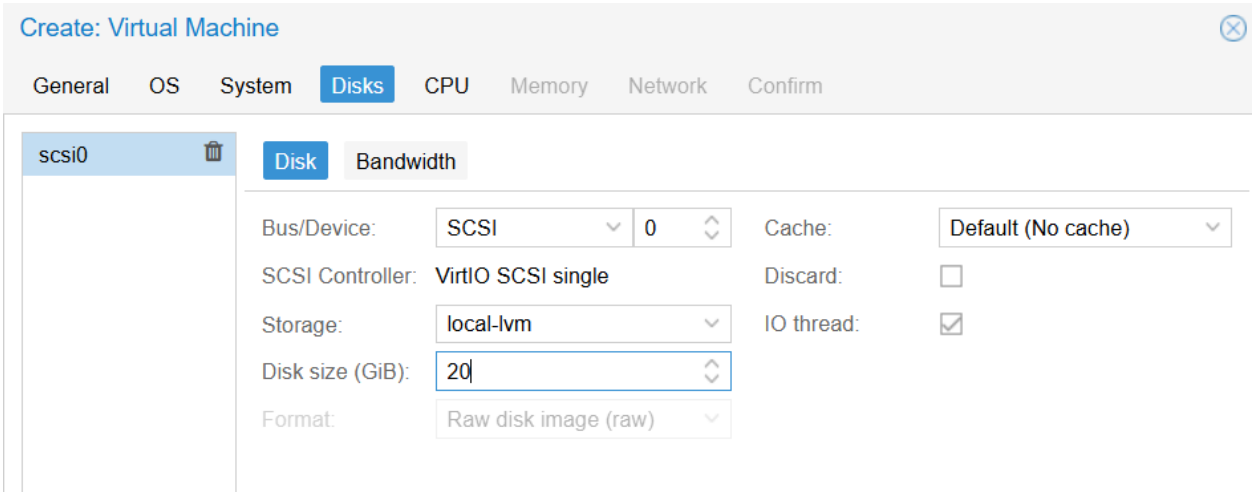
2.Скачиваем ВМ на proxmox



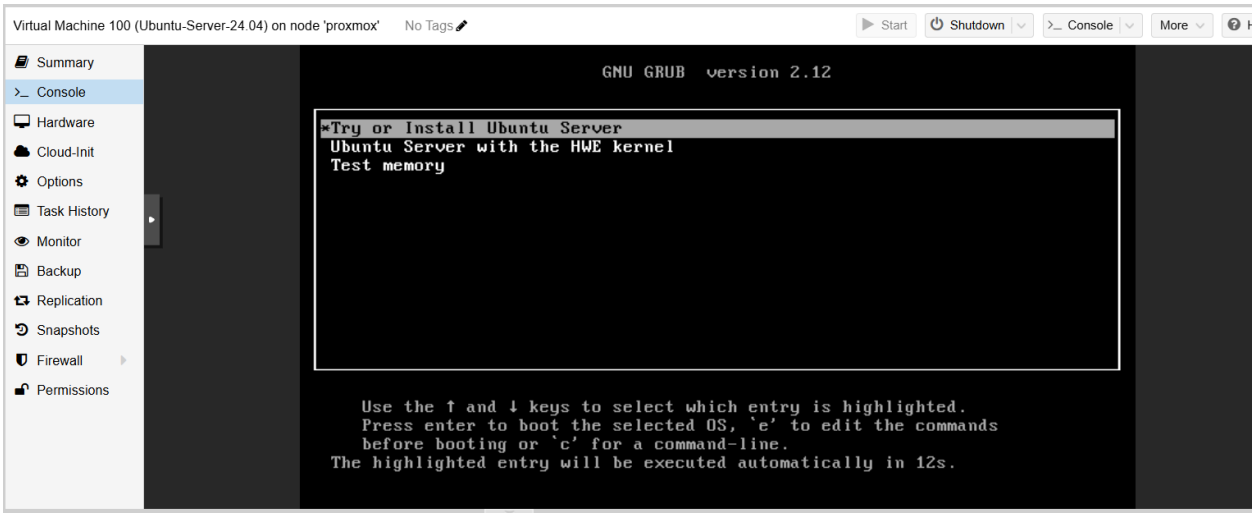








Начинаем установку

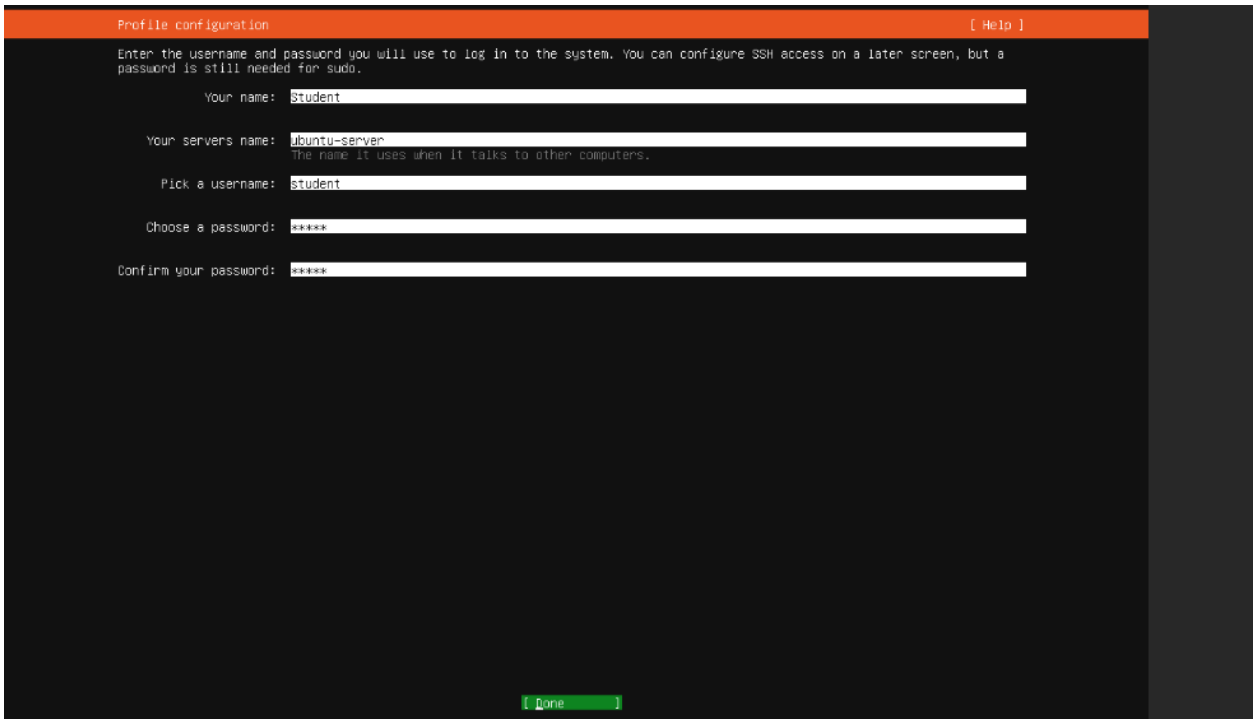


Далее я просто заполняю данные ВМ

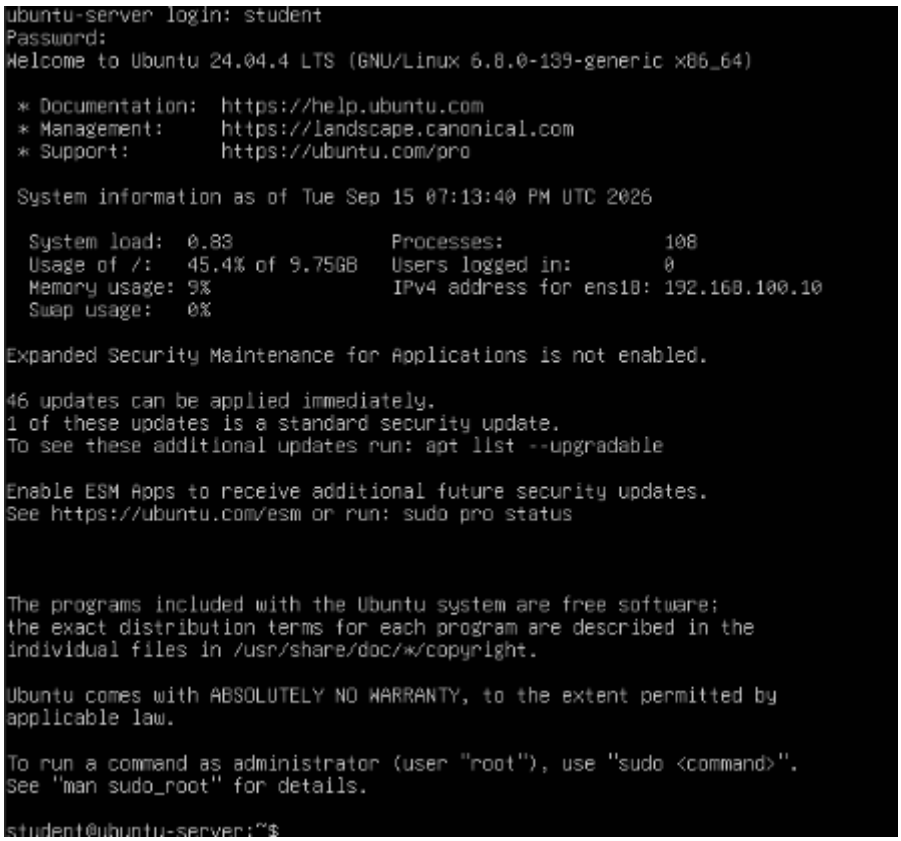
2 ядра CPU

2048 МБ RAM

20 ГБ диск



В итоге у меня получилось всё скачать



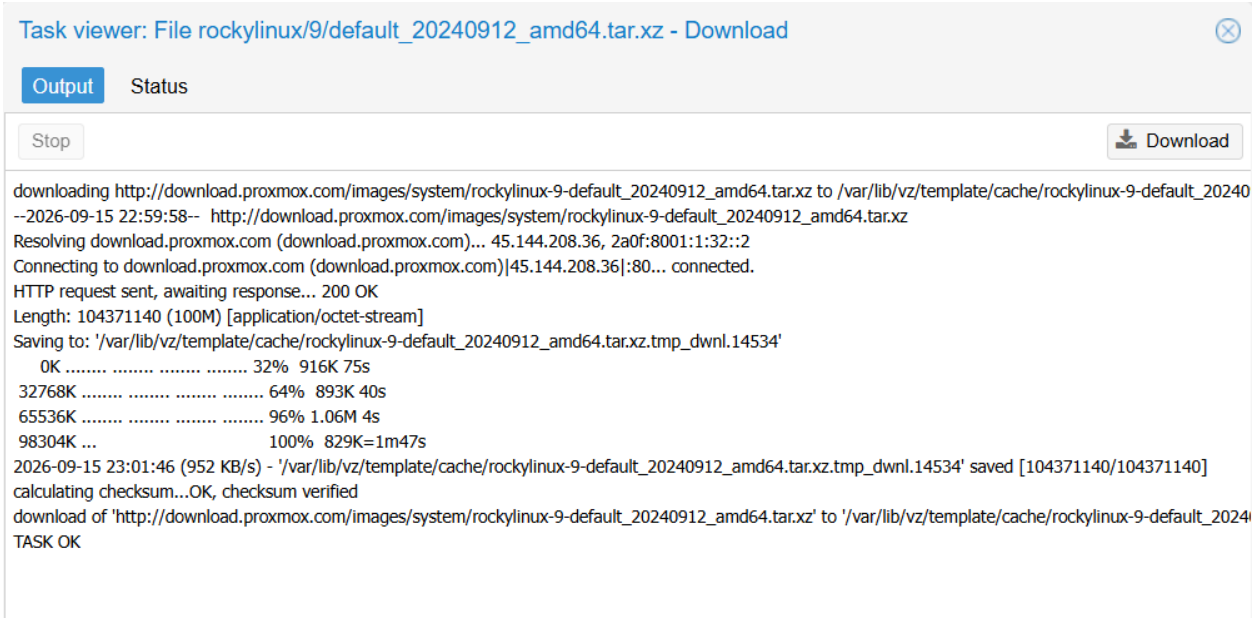
Теперь проверяем всё ли работает

```
student@ubuntu-server:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: ens18: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether bc:24:11:10:36:b2 brd ff:ff:ff:ff:ff:ff
    altname enp0s18
    inet 192.168.100.10/24 brd 192.168.100.255 scope global ens18
       valid_lft forever preferred_lft forever
    inet6 fe80::be24:11ff:fe10:36b2/64 scope link
       valid_lft forever preferred_lft forever
student@ubuntu-server:~$ ping 8.8.8.8
PING 8.8.8.8 (8.8.8.8) 56(84) bytes of data.
64 bytes from 8.8.8.8: icmp_seq=1 ttl=254 time=34.1 ms
64 bytes from 8.8.8.8: icmp_seq=2 ttl=254 time=49.6 ms
64 bytes from 8.8.8.8: icmp_seq=3 ttl=254 time=155 ms
^C
--- 8.8.8.8 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2011ms
rtt min/avg/max/mdev = 34.118/79.632/155.219/53.818 ms
student@ubuntu-server:~$ ping ya.ru
PING ya.ru (77.88.55.242) 56(84) bytes of data.
64 bytes from ya.ru (77.88.55.242): icmp_seq=2 ttl=254 time=46.5 ms
64 bytes from ya.ru (77.88.55.242): icmp_seq=3 ttl=254 time=27.5 ms
64 bytes from ya.ru (77.88.55.242): icmp_seq=4 ttl=254 time=21.6 ms
^C
--- ya.ru ping statistics ---
4 packets transmitted, 3 received, 25% packet loss, time 3041ms
rtt min/avg/max/mdev = 21.577/31.865/46.470/10.611 ms
```

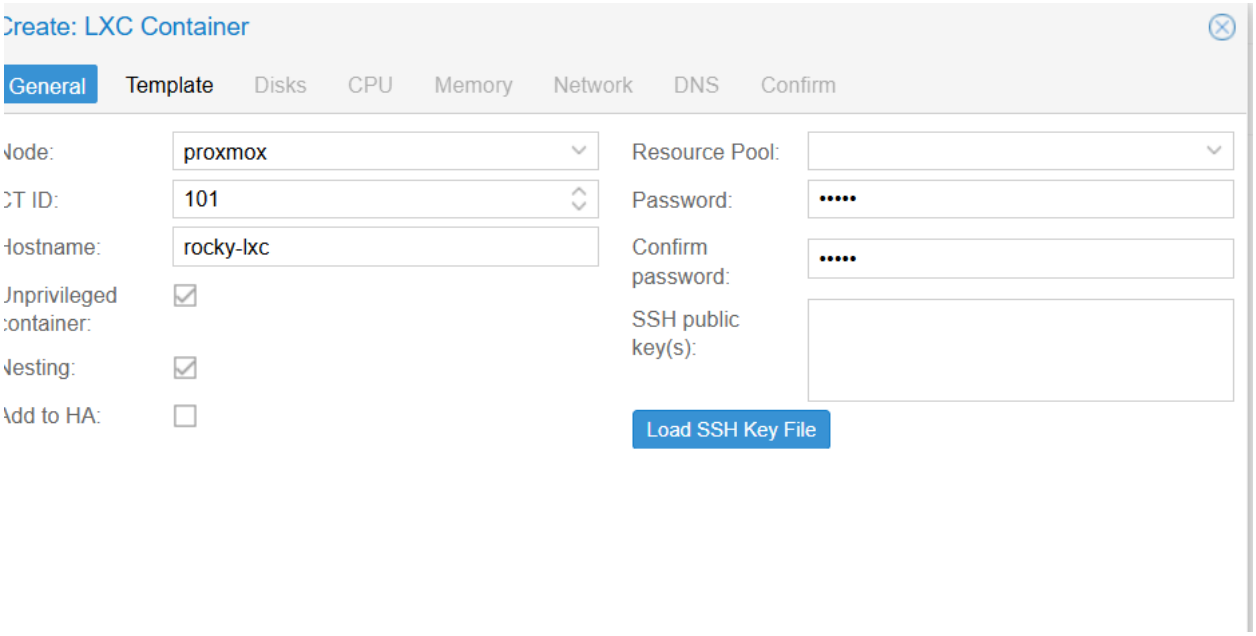
Задание №5 Создание контейнера LXC

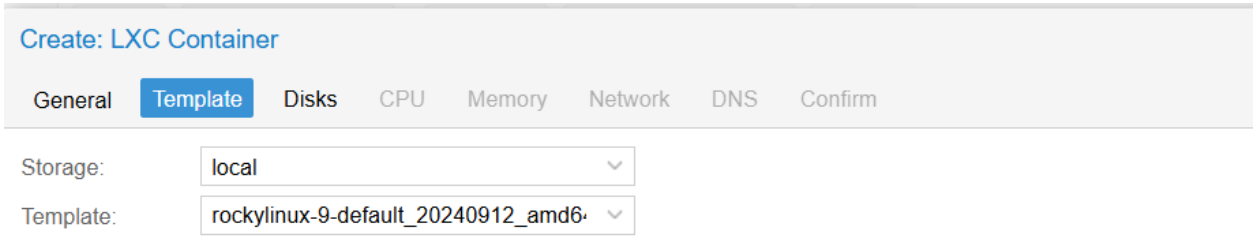
Загрузка шаблона

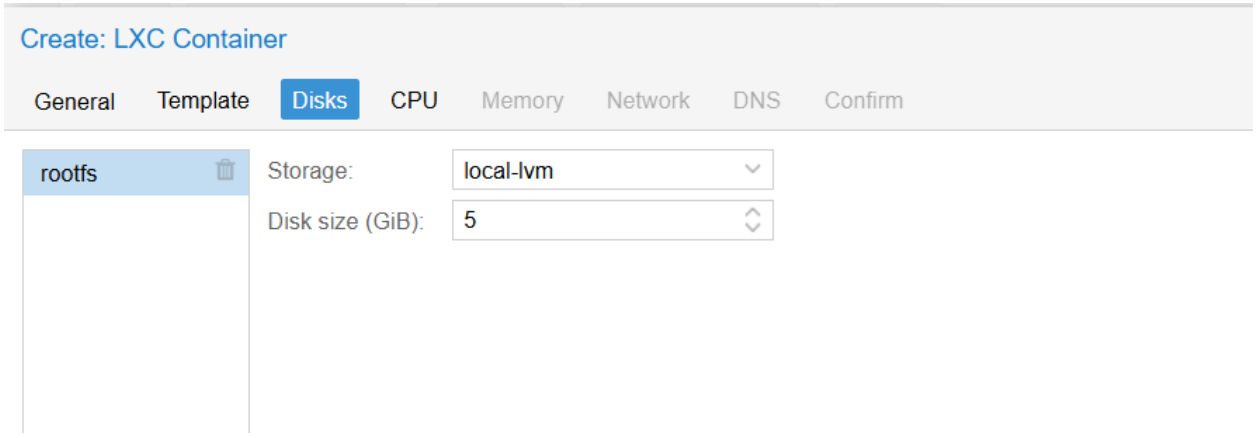
Скачиваем rockylinux

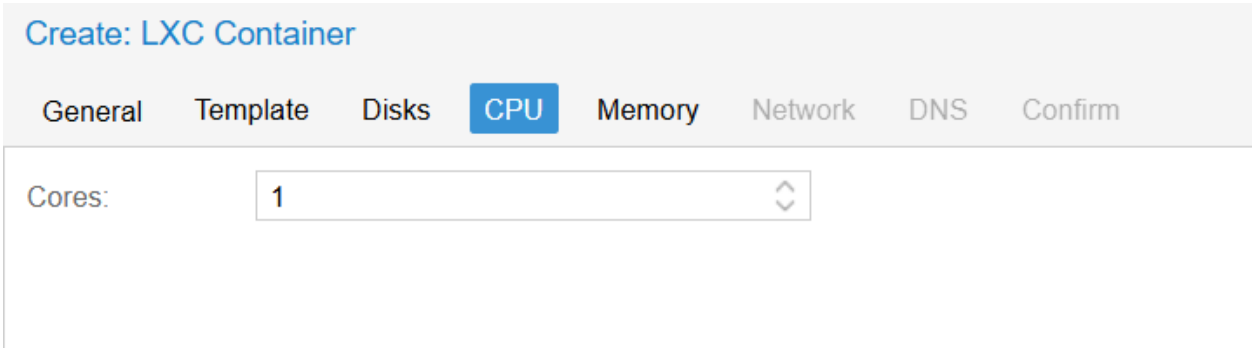


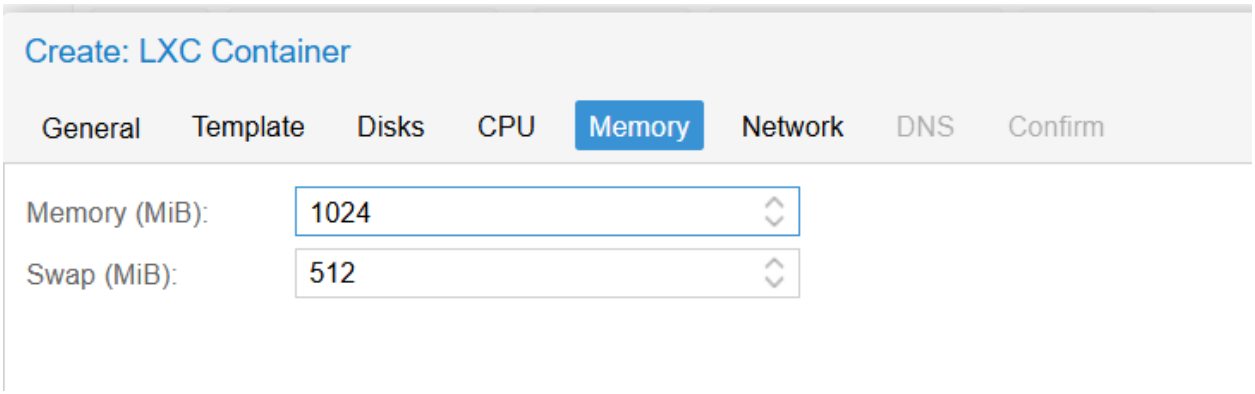
Теперь создаём контейнер

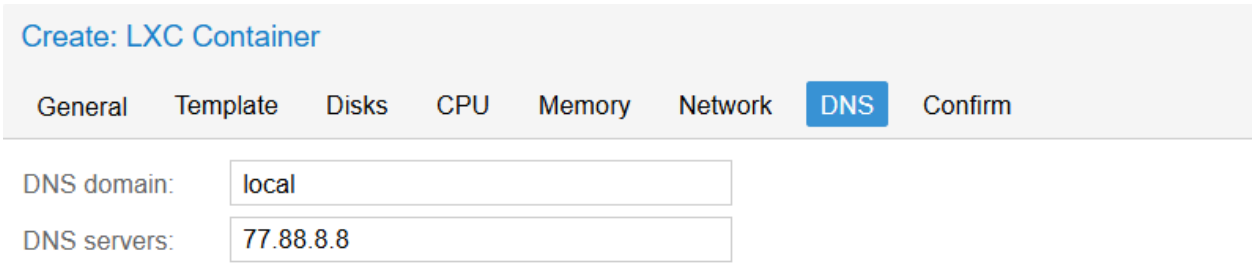

Create: LXC Container
General
Template
Disks
CPU
Memory
Network
DNS
Confirm
Node:
proxmox
CT ID:
101
Hostname:
rocky-lxc
Unprivileged container:
Nesting:
Add to HA:
Resource Pool:
Password:
Confirm password:
SSH public key(s):
Load SSH Key File


Create: LXC Container
General
Template
Disks
CPU
Memory
Network
DNS
Confirm
Storage:
local
Template:
rockylinux-9-default_20240912_amd64


Create: LXC Container
General
Template
Disks
CPU
Memory
Network
DNS
Confirm
rootfs
Storage:
local-lvm
Disk size (GiB):
5


Create: LXC Container
General
Template
Disks
CPU
Memory
Network
DNS
Confirm
Cores:
1


Create: LXC Container
General
Template
Disks
CPU
Memory
Network
DNS
Confirm
Memory (MiB):
1024
Swap (MiB):
512


Create: LXC Container
General
Template
Disks
CPU
Memory
Network
DNS
Confirm
DNS domain:
local
DNS servers:
77.88.8.8

Create: LXC Container

General | Template | Disks | CPU | Memory | Network | DNS | **Confirm**

Key ↑	Value
cores	1
features	nesting=1
hostname	rocky-lxc
memory	1024
nameserver	77.88.8.8
net0	name=eth0,bridge=vmbr1,firewall=1,ip=192.168.100.20/24,gw=192.168.100.1
nodename	proxmox
ostemplate	local:vztmpl/rockylinux-9-default_20240912_amd64.tar.xz
pool	
rootfs	local-lvm:5
searchdomain	local
ssh-public-keys	
swap	512
unprivileged	1
vmid	101

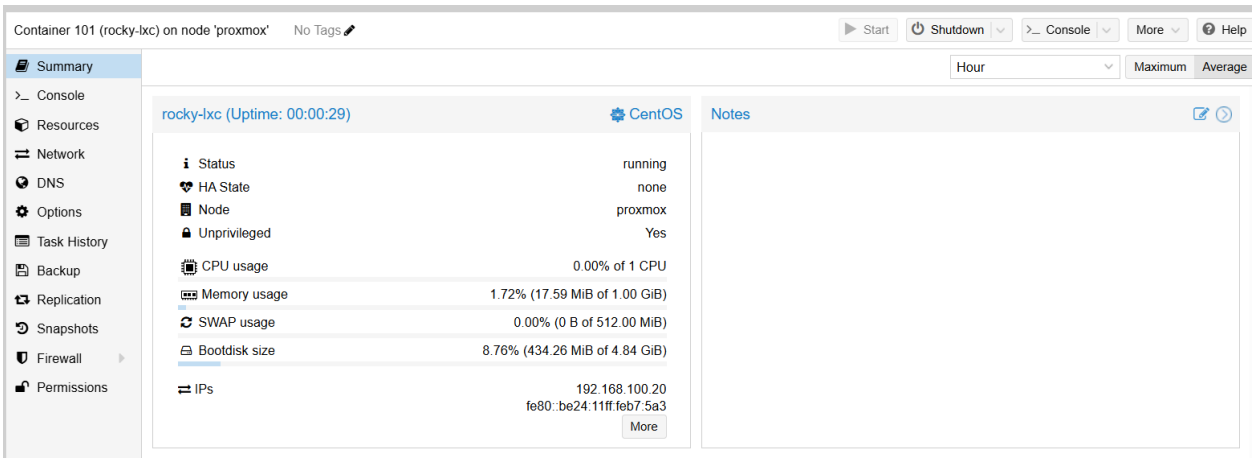
Task viewer: CT 101 - Create

Output | Status

Stop

```
  WARNING: You have not turned on protection against thin pools running out of space.
  WARNING: Set activation/thin_pool_autoextend_threshold below 100 to trigger automatic extension of thin pools before they get full.
  Logical volume "vm-101-disk-0" created.
  WARNING: Sum of all thin volume sizes (25.00 GiB) exceeds the size of thin pool pve/data and the amount of free space in volume group (<6.13
  Logical volume pve/vm-101-disk-0 changed.
Creating filesystem with 1310720 4k blocks and 327680 inodes
Filesystem UUID: b194be4d-1ce3-4a18-b2aa-a1bda2e70194
Superblock backups stored on blocks:
        32768, 98304, 163840, 229376, 294912, 819200, 884736
extracting archive '/var/lib/vz/template/cache/rockylinux-9-default_20240912_amd64.tar.xz'
Total bytes read: 446238720 (426MiB, 78MiB/s)
Detected container architecture: amd64
Creating SSH host key 'ssh_host_rsa_key' - this may take some time ...
done: SHA256:D4JzgHfR6S9RkLePg1eF5FTUp2/bRoq2fyUrfrNe+88 root@rocky-lxc
Creating SSH host key 'ssh_host_ed25519_key' - this may take some time ...
done: SHA256:g+RgXM2RwhnJeq8tSUCzOu/CyxxcZpdmPlhVyVY223w root@rocky-lxc
Creating SSH host key 'ssh_host_ecdsa_key' - this may take some time ...
done: SHA256:xG71LYgK7jQOwIcHIOC1inTI8vjNwe5MmssYkfXLHM0 root@rocky-lxc
TASK OK
```

Запуск и проверка




```
Rocky Linux 9.4 (Blue Onyx)
Kernel 7.0.14-17-pve on an x86_64

rocky-lxc login: root
Password:
[root@rocky-lxc ~]# ping 8.8.8.8
PING 8.8.8.8 (8.8.8.8) 56(84) bytes of data.
64 bytes from 8.8.8.8: icmp_seq=1 ttl=254 time=27.2 ms
64 bytes from 8.8.8.8: icmp_seq=2 ttl=254 time=27.1 ms
64 bytes from 8.8.8.8: icmp_seq=3 ttl=254 time=57.3 ms

--- 8.8.8.8 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2007ms
rtt min/avg/max/mdev = 27.094/37.200/57.315/14.223 ms
[root@rocky-lxc ~]# ping ya.ru
PING ya.ru (77.88.44.242) 56(84) bytes of data.
64 bytes from ya.ru (77.88.44.242): icmp_seq=1 ttl=254 time=7.30 ms
64 bytes from ya.ru (77.88.44.242): icmp_seq=2 ttl=254 time=11.3 ms

--- ya.ru ping statistics ---
2 packets transmitted, 2 received, 0% packet loss, time 1020ms
rtt min/avg/max/mdev = 7.297/9.284/11.271/1.987 ms
[root@rocky-lxc ~]#
```